(502) 472-5677 **Karthik Nayak** **karthiknayak@uky.edu**

Education

Bachelor of Science in Computer Science — University of Kentucky
GPA: 3.66 / 4.0, Junior — Projected graduation: May 19
Relevant Courses: Algorithm Design & Analysis, Machine Learning, Interactive Machine Learning, Graphics and Multimedia (Java), Embedded Systems, Program Design & Abstraction, Discrete Math

Skills

Programming	Concepts	Automation	Front-End
Python	Machine Learning	PLC programming	JavaScript
Java	Image Processing	HMI design (UI)	HTML
C++	API Design for SaaS	Mechatronics	CSS
C	OOP	AutoCAD Electrical	Graphic Design

GitHub: /karthiknayak02 **Website:** karthiknayak02.github.io **Proficient Languages:** English, Konkani, Hindi, and Marathi

Work Experience

IBM *June. 2018 – Present*

Software Developer Intern *Dallas, TX*

- Worked on IBM's Business Transaction Intelligence SaaS product to convert existing monolithic BTI code to microservice architecture
- Designed and developed several REST API's in JAVA which supports the analytics section of the BTI service
- Deployed and tested project implementation on Docker images and IBM liberty servers
- Worked in an agile/scrum software development environment in a team of three

University of Kentucky, Computer Science department *Oct. 2017 – May. 2018*

Undergraduate Research Assistant *Lexington, KY*

- Worked under Dr. James Griffioen and Dr. Zongming Fei's on an NSF-funded research project to study and test the implementation of Software-Defined Networking (SDN) on campus network architecture.
- Created a client/server storage application to make use of special campus VIP internet lanes to achieve faster speeds. Built using Python along with Flask web framework to support REST APIs.
- Conducted and examined speed tests of SDN's through wireless access points using rclone.
- Interpreted DNS responses and TCP transfers of packets through campus network using Wireshark & UNIX tools (tcpdump)

Tesla *May. 2017 – Aug. 2017*

Automation Controls Engineering Intern *Gigafactory 1, Reno, NV*

- Worked in the Model 3 Controls Team to support software design and controls programming of several machines on the production line.
- Designed user interface (UI) templates and integrated PLC programs in solo and team projects on the powertrain.
- Developed PLC Add-on Instructions (AOI) packages; for interactions between Automated Guided Vehicles (AGV) and workstation conveyers. Also to handle simultaneous release of two back-to-back conveyor stoppers to increase production cycle time.
- Responsible for testing AGV-to-machine interfaces at GF1 machines by preparing and using a simulator to identify and correct programming flaws in the equipment ahead of the actual AGV arrival, also to debug AGV itself.
- Assisted four interns from Material Flow Engineering Team to assemble electrical cabinets housing a PLC, lights/buttons, an Optical Data Transmission Device for the AGV simulation benches mentioned above.
- Supported safety buyoffs (prerequisite checks) at Model 3 Drive Unit Final Assembly.

Visumatic Industrial Product *May. 2016 – Aug. 2016*

Computer & Electrical Engineering Intern *Lexington, KY*

Visumatic is an OEM of fastener "feed and drive" equipment and automated systems.

- Gained first-hand knowledge in the development and implementation of automated systems for manufacturing.
- Responsible for programming a dual driver machine which feeds and drives screws into safety glasses.
- Programmed a machine with a pneumatic driver, touch HMI display and a rotatory indexing system.
- Programmed a feeder system with dual Siamese twins' fastener outfeed.
- Supported the calibration and build of 6-Axis Robot Systems; helped configure vision systems, OS, troubleshoot electro-pneumatically controlled mechanical feeders and drivers, verify electrical schematics, electrical wiring.

Work Experience

2nd Place, Cathacks national college hackathon (Major League Hacking) *2018*

- Developed a YouTube Chrome Extension which applies Natural Language Processing (NLP) on video captions to extract interactive topics in the video. When clicked uses code injection to automate video scrubbing to the corresponding section of the video (Team of 4)

2nd Place, Cathacks national college hackathon & winner of AWS challenge (Major League Hacking) *2017*

- Developed an Amazon Alexa application with AWS to get user speech input to generate 3D Braille STL files (Team of 2)

Eta Kappa Nu – ECE Honor Society for students in upper 25% of the junior class of electrical and computer engineering majors *2017*